Data Analytics (DA)
Module -2
Power BI Assignment 2 – DAX & Data Visualization

Tasks

DAX Calculations

Calculated Columns

Create a Calculated Column for 'Category Type':

```
1 Category Type = 'Order Details'[Category]& "-"&'Order Details'[Sub-Category]
```

Order ID	Amount	Profit	Quantity	Category	Sub-Category	Profit Margin	Profit status	Category Type
B-25603	12	1	2	Clothing	Hankerchief	0.083333333	Profit	Clothing-Hankerchief
B-25608	257	23	5	Clothing	Hankerchief	0.089494163	Profit	Clothing-Hankerchief
B-25615	68	20	5	Clothing	Hankerchief	0.294117647	Profit	Clothing-Hankerchief
B-25616	42	12	5	Clothing	Hankerchief	0.285714286	Profit	Clothing-Hankerchief
B-25624	26	12	3	Clothing	Hankerchief	0.461538462	Profit	Clothing-Hankerchief
B-25625	97	29	2	Clothing	Hankerchief	0.298969072	Profit	Clothing-Hankerchief
B-25635	40	16	3	Clothing	Hankerchief	0.4	Profit	Clothing-Hankerchief
B-25638	154	39	3	Clothing	Hankerchief	0.253246753	Profit	Clothing-Hankerchief
B-25654	34	12	3	Clothing	Hankerchief	0.352941176	Profit	Clothing-Hankerchief
B-25656	6	3	1	Clothing	Hankerchief	0.5	Profit	Clothing-Hankerchief
B-25656	56	18	2	Clothing	Hankerchief	0.321428571	Profit	Clothing-Hankerchief
B-25670	24	1	2	Clothing	Hankerchief	0.041666667	Profit	Clothing-Hankerchief
B-25670	14	2	1	Clothing	Hankerchief	0.142857143	Profit	Clothing-Hankerchief
B-25730	18	8	2	Clothing	Hankerchief	0.444444444	Profit	Clothing-Hankerchief

Calculate Revenue per Order in Order Details Table

```
1 Revenue = 'Order Details'[Amount]*'Order Details'[Quantity]
```

Order ID	Amount	Profit	Quantity	Category	Sub-Category	Profit Margin	Profit status	Category Type	Revenue
B-25603	12	1	2	Clothing	Hankerchief	0.083333333	Profit	Clothing-Hankerchief	24
B-25608	257	23	5	Clothing	Hankerchief	0.089494163	Profit	Clothing-Hankerchief	1285
B-25615	68	20	5	Clothing	Hankerchief	0.294117647	Profit	Clothing-Hankerchief	340
B-25616	42	12	5	Clothing	Hankerchief	0.285714286	Profit	Clothing-Hankerchief	210
B-25624	26	12	3	Clothing	Hankerchief	0.461538462	Profit	Clothing-Hankerchief	78
B-25625	97	29	2	Clothing	Hankerchief	0.298969072	Profit	Clothing-Hankerchief	194
B-25635	40	16	3	Clothing	Hankerchief	0.4	Profit	Clothing-Hankerchief	120
B-25638	154	39	3	Clothing	Hankerchief	0.253246753	Profit	Clothing-Hankerchief	462
B-25654	34	12	3	Clothing	Hankerchief	0.352941176	Profit	Clothing-Hankerchief	102

Create a Calculated Column to Categorize Sales

```
1 Sales_Category = IF('Order Details'[Amount]>CALCULATE(AVERAGE('Order Details'[Amount]),ALL('Order Details')), "Above Average ", "below Average")
```

Amount	Profit	Quantity	Category	Sub-Category	Profit Margin	Profit status	Category Type	Revenue	Sales_Category
12	1	2	Clothing	Hankerchief	0.083333333	Profit	Clothing-Hankerchief	24	below Average
257	23	5	Clothing	Hankerchief	0.089494163	Profit	Clothing-Hankerchief	1285	below Average
68	20	5	Clothing	Hankerchief	0.294117647	Profit	Clothing-Hankerchief	340	below Average
42	12	5	Clothing	Hankerchief	0.285714286	Profit	Clothing-Hankerchief	210	below Average
26	12	3	Clothing	Hankerchief	0.461538462	Profit	Clothing-Hankerchief	78	below Average
97	29	2	Clothing	Hankerchief	0.298969072	Profit	Clothing-Hankerchief	194	below Average

Calculated Measures

1. Order Count-

```
1 Order_count = COUNT('Order Details'[Order ID])
```

Average_orderdetail_amount	Order_count	Average_Profit
287.67	2K	15.97

Average Profit in Delhi-

```
1 Average_profit_delhi = CALCULATE(AVERAGE('Order Details'[Profit]),'List of Orders'[City] ="Delhi")
```

2K	15.97	43.31	158.80K

Year-to-Date (YTD) Sales-

```
1 YTD_sales = TOTALYTD([Total_Sales],'List of Orders'[Order Date])
```

2K	15.97	43.31	158.80K

Data Visualization

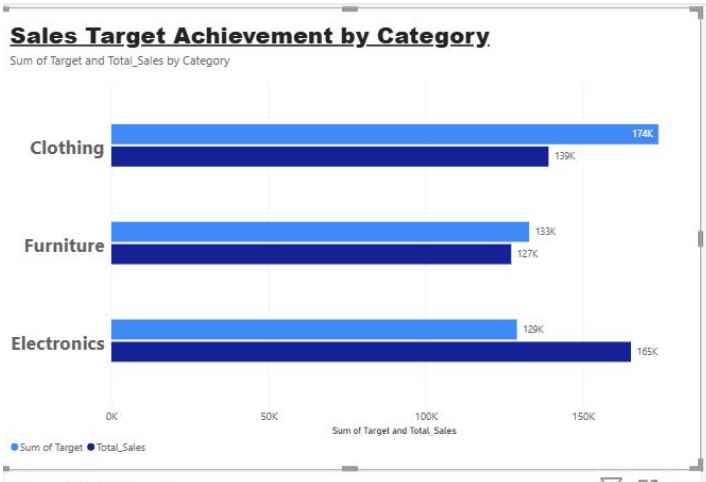
Create the following visualizations using Power BI:

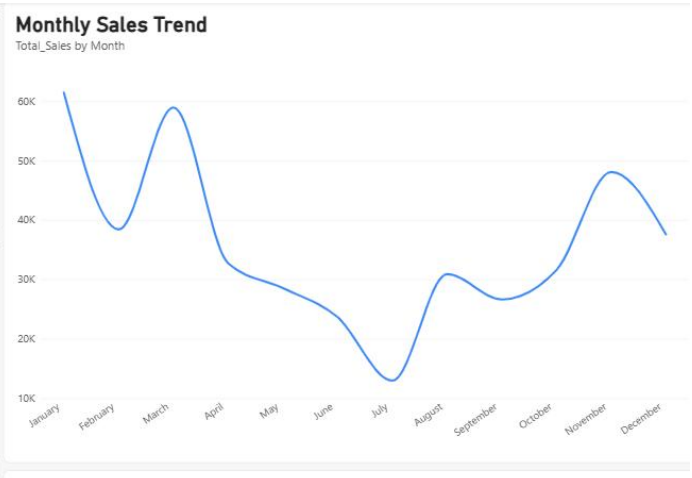
Sales Target Achievement by Category:

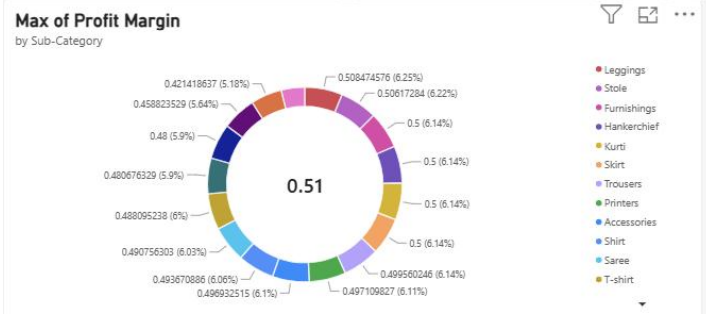
Max Profit Margin by Sub-Category

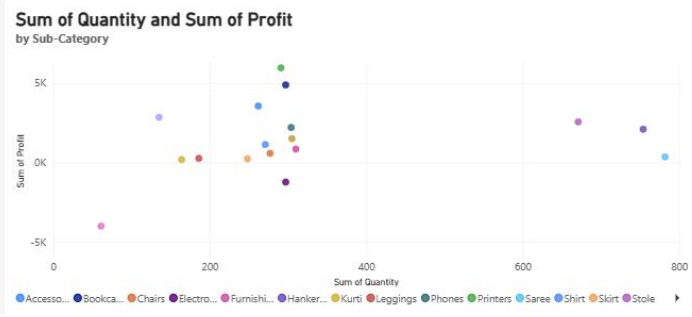
Monthly Sales

Comparison of Profit and Quantity by Sub-Category:









Comparison of Total Sales Amount and Target

Category	Min_Target
Clothing	12,000.00
Electronics	9,000.00
Furniture	10,400.00

```
1  Sales vs Taget = [Total_Sales]-[Total_target]
```

	2K	15.97	43.31	158.80K	0.51

```
1  Total_Sales = SUM('Order Details'[Amount])
```

	2K	15.97
Magin_Subcat	Total_Sales 431.50K	Total_target 435.90K

```
1  Total_target = sum('Sales target'[Target])
```

	2K	15.97	43.31
gin_Subcat	Total_Sales 431.50K	Total_target 435.90K	Min_Target 9.00K

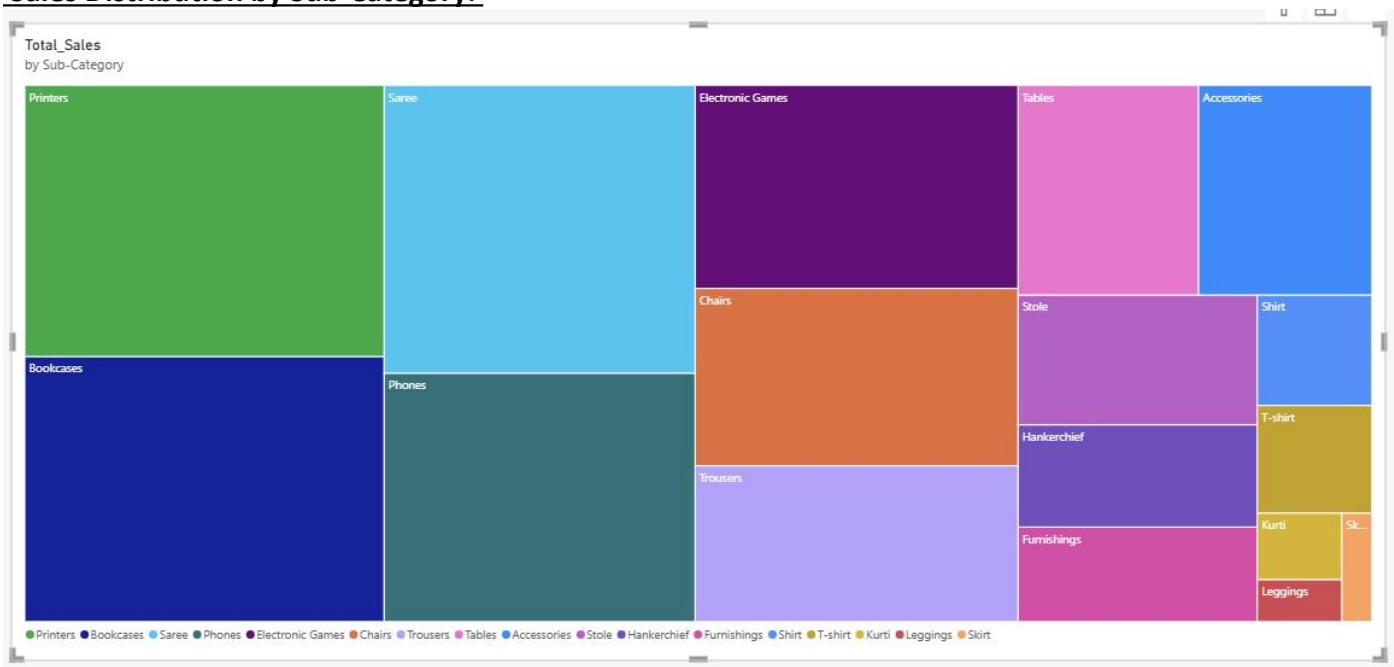
Sales Performance Matrix

Category	January	February	March	April	May	June	July	August	September	October	November	December	Total
Clothing	139,054.00	139,054.00	139,054.00	139,054.00	139,054.00	139,054.00	139,054.00	139,054.00	139,054.00	139,054.00	139,054.00	139,054.00	139,054.00
Electronics	165,267.00	165,267.00	165,267.00	165,267.00	165,267.00	165,267.00	165,267.00	165,267.00	165,267.00	165,267.00	165,267.00	165,267.00	165,267.00
Furniture	127,181.00	127,181.00	127,181.00	127,181.00	127,181.00	127,181.00	127,181.00	127,181.00	127,181.00	127,181.00	127,181.00	127,181.00	127,181.00
Total	431,502.00	431,502.00	431,502.00	431,502.00	431,502.00	431,502.00	431,502.00	431,502.00	431,502.00	431,502.00	431,502.00	431,502.00	431,502.00

Geographic Sales Analysis



Sales Distribution by Sub-Category:



Order Count Analysis by State:

